

THE SWISS-CHEESE OPERAD $\mathbf{SC}^{\text{ch},\text{top}}$ AND PVA DESCENT

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ABSTRACT. A chiral algebra $\mathcal{A}$ on a smooth curve X determines a bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\mathcal{A})$ that is an $\mathbf{E}_1$ chiral coassociative coalgebra: the differential encodes holomorphic factorisation on $\text{FM}_k(\mathbb{C})$, the deconcatenation coproduct encodes topological factorisation on $\text{Conf}_k^<(\mathbb{R})$. The two-coloured operad governing the interaction of these two structures on the *derived chiral centre* $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ and the boundary algebra $\mathcal{A}$ is the Swiss-cheese operad $\mathbf{SC}^{\text{ch},\text{top}}$, with closed colour $\text{FM}_k(\mathbb{C})$, open colour $\mathbf{E}_1(m) = \text{Conf}_m^<(\mathbb{R})$, and empty open-to-closed component (directionality).

We establish three results. First, $\mathbf{SC}^{\text{ch},\text{top}}$ is homotopy-Koszul, via Kontsevich formality for the closed colour composed with Livernet’s Koszulity of the classical Swiss-cheese operad $\mathbf{SC}$ and bar-cobar transfer. Second, the Koszul dual cooperad satisfies $(\mathbf{SC}^{\text{ch},\text{top}})^! = (\text{Lie}^c, \text{Ass}^c, \text{shuffle-mixed})$; in particular $\mathbf{SC}^{\text{ch},\text{top}}$ is *not* Koszul self-dual ($\dim \mathbf{SC}_{\text{cl}}^{\text{ch},\text{top}}(n) = 1$ versus $\dim(\mathbf{SC}^{\text{ch},\text{top}})^!_{\text{cl}}(n) = (n-1)!$). Third, the cogenerator projections of the bar differential produce A_∞ operations $\{m_k\}_{k \geq 1}$ whose Stasheff relations are Stokes’ theorem on $\text{FM}_n(\mathbb{C})$; on cohomology these descend to the five axioms of a Poisson vertex algebra. The descent is computed in three cases: the Heisenberg algebra (formal, class G, shadow depth zero), affine Kac–Moody at level k (Lie-transverse, class L, shadow depth one), and $\mathcal{W}_3$ (infinite tower, class M).

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1. INTRODUCTION: TWO COLOURS, ONE OPERAD

1.1. **The problem.** The bar complex of a chiral algebra carries two independent structures. The *differential* $D_{\mathcal{A}}$ is a coderivation of the cofree tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$; it encodes the OPE residues extracted from collisions on the Fulton–MacPherson spaces $\text{FM}_k(\mathbb{C})$. The *deconcatenation coproduct* Δ cuts the tensor word at each gap:

$$\Delta[s^{-1}a_1 \mid \cdots \mid s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 \mid \cdots \mid s^{-1}a_i] \otimes [s^{-1}a_{i+1} \mid \cdots \mid s^{-1}a_n]. \quad (1.1)$$

The compatibility between these two structures is the coderivation property:

$$\Delta \circ D_{\mathcal{A}} = (D_{\mathcal{A}} \otimes \text{id} + \text{id} \otimes D_{\mathcal{A}}) \circ \Delta. \quad (1.2)$$

A differential and a coassociative coproduct satisfying (1.2) make $(\bar{B}(\mathcal{A}), D_{\mathcal{A}}, \Delta)$ into a dg coassociative coalgebra: an E_1 coalgebra, with holomorphic collisions on $\text{FM}_k(\mathbb{C})$ as the differential and topological separations on $\text{Conf}_k^<(\mathbb{R})$ as the coproduct.

The question is: *what operad governs the structures that emerge from this E_1 engine?*

1.2. **The answer.** The bar complex itself does not carry the answer. It is the input, not the output. The output is the *chiral derived centre*: the chiral Hochschild cochain complex $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$, defined via the chiral endomorphism

operad $\text{End}_{\mathcal{A}}^{\text{ch}}$ with spectral parameters from $\text{FM}_k(\mathbb{C})$. The pair

$$(C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}), \mathcal{A}) = (\text{bulk}, \text{boundary}) \quad (1.3)$$

is the datum governed by the two-coloured Swiss-cheese operad $\text{SC}^{\text{ch,top}}$. The closed colour is the bulk; the open colour is the boundary. Bulk acts on boundary; boundary does not act on bulk. This directionality is the empty open-to-closed component of $\text{SC}^{\text{ch,top}}$.

Three properties hold.

- (i) **Homotopy-Koszulity.** The operad $\text{SC}^{\text{ch,top}}$ is homotopy-Koszul (Theorem 3.3): the bar-cobar adjunction $\Omega\mathbf{B}(\text{SC}^{\text{ch,top}}) \xrightarrow{\sim} \text{SC}^{\text{ch,top}}$ is a quasi-isomorphism. This follows from Livernet's Koszulity of the classical Swiss-cheese operad SC [13], composed with Kontsevich formality for the closed colour, and transferred to $\text{SC}^{\text{ch,top}}$ via bar-cobar comparison.
- (ii) **The Koszul dual cooperad.** The Koszul dual satisfies $(\text{SC}^{\text{ch,top}})^! = (\text{Lie}^c, \text{Ass}^c, \text{shuffle-mixed})$ (Theorem 4.1). The closed colour exchanges $\text{Com} \leftrightarrow \text{Lie}$; the open colour preserves Ass (which is self-dual); the mixed sector acquires a shuffle structure. In particular, $\text{SC}^{\text{ch,top}}$ is *not* Koszul self-dual: the closed-colour component dimensions differ ($\dim \text{SC}_{\text{cl}}^{\text{ch,top}}(n) = 1$ versus $\dim (\text{SC}^{\text{ch,top}})^!_{\text{cl}}(n) = (n-1)!$).
- (iii) **PVA descent.** The cogenerator projections of $D_{\mathcal{A}}$ yield A_{∞} operations $\{m_k\}_{k \geq 1}$ satisfying the Stasheff relations, each relation being Stokes' theorem on $\text{FM}_n(\mathbb{C})$. On cohomology, the operations m_k descend to a commutative product and a λ -bracket satisfying the five axioms of a Poisson vertex algebra: sesquilinearity, skew-symmetry, Jacobi, Leibniz, and commutativity (Theorem 5.3).

Warning 1.1 (The bar complex is not an $\text{SC}^{\text{ch,top}}$ -coalgebra). The bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\overline{\mathcal{A}})$ is an E_1 chiral coassociative coalgebra. It carries a differential (from the chiral product) and a coproduct (from deconcatenation). These are two structures on a *single* coalgebra. The $\text{SC}^{\text{ch,top}}$ structure is *not* carried by $\overline{B}(\mathcal{A})$; it is carried by the pair $(C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}), \mathcal{A})$, where $C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$ is computed *using* the bar complex as a resolution. The bar complex is the E_1 engine; the derived centre pair is the $\text{SC}^{\text{ch,top}}$ output.

1.3. Context and conventions. We work over a field $\mathbb{k}$ of characteristic zero. Grading is cohomological: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal: $\overline{B}(\mathcal{A}) = T^c(s^{-1}\overline{\mathcal{A}})$ where $\overline{\mathcal{A}} = \ker(\varepsilon)$. The Fulton–MacPherson space $\text{FM}_n(X)$ is the iterated real blowup of X^n along all diagonal strata. Configuration spaces of ordered points on $\mathbb{R}$ are denoted $\text{Conf}_n^<(\mathbb{R}) = \{t_1 < \cdots < t_n\}$.

Throughout, $\mathcal{A}$ denotes a chiral algebra on a smooth algebraic curve X over $\mathbb{k}$ in the sense of Beilinson–Drinfeld [2]. A vertex algebra is a chiral algebra on the formal disk $D = \text{Spec } \mathbb{k}[[z]]$.

1.4. Relation to other work. The classical Swiss-cheese operad $\mathbf{SC}$ (with $\mathbf{E}_2$ closed colour and $\mathbf{E}_1$ open colour) was introduced by Voronov [16]. Koszulity of $\mathbf{SC}$ was proved by Livernet [13]. The deformation-theoretic role of $\mathbf{SC}$ in open-closed homotopy algebras was developed by Kajiura–Stasheff [12]. The passage from the classical $\mathbf{SC}$ to the chiral variant $\mathbf{SC}^{\text{ch},\text{top}}$ replaces the $\mathbf{E}_2$ closed colour by $\mathbf{FM}_k(\mathbb{C})$ -spaces carrying holomorphic structure; this replacement is the content of the Beilinson–Drinfeld factorisation framework [2]. The relationship between A_∞ structures and PVA axioms has been studied from the λ -bracket perspective by De Sole and Kac [5], and from the factorisation perspective by Francis and Gaitsgory [6]. The present paper makes the operadic architecture of this descent explicit.

2. THE $\mathbf{SC}^{\text{ch},\text{top}}$ OPERAD

2.1. Configuration spaces.

Definition 2.1 (Fulton–MacPherson space). For a smooth manifold M and $n \geq 1$, the *Fulton–MacPherson space* $\mathbf{FM}_n(M)$ is the real-oriented blowup of M^n along all diagonal strata $\Delta_S = \{(x_i)_{i \in I} : x_i = x_j \text{ for all } i, j \in S\}$, for all subsets $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$, performed in order of increasing $|S|$. Points of $\mathbf{FM}_n(M)$ are configurations of n labelled points in M , together with “infinitesimal data” recording the relative rates and directions of approach to collision.

Definition 2.2 (Ordered configuration space). For $n \geq 1$, define

$$\text{Conf}_n^<(\mathbb{R}) = \{(t_1, \dots, t_n) \in \mathbb{R}^n : t_1 < t_2 < \dots < t_n\}.$$

This is contractible: $\text{Conf}_n^<(\mathbb{R}) \cong \mathbb{R}_{>0}^{n-1}$ via the change of variables $u_i = t_{i+1} - t_i$.

2.2. The two-coloured operad.

Definition 2.3 (The Swiss-cheese operad $\mathbf{SC}^{\text{ch},\text{top}}$). The operad $\mathbf{SC}^{\text{ch},\text{top}}$ is a two-coloured operad with colours $\{\text{cl}, \text{op}\}$ (closed and open), defined by the following operation spaces:

- (i) **Closed-to-closed.** For $k \geq 1$ closed inputs and one closed output:

$$\mathbf{SC}^{\text{ch},\text{top}}(\text{cl}^k; \text{cl}) = \mathbf{FM}_k(\mathbb{C}). \quad (2.1)$$

Composition is given by the operadic structure maps of the Fulton–MacPherson operad: for a decomposition $\{1, \dots, k\} = S_1 \sqcup \dots \sqcup S_r$, the composition map

$$\gamma: \mathbf{FM}_r(\mathbb{C}) \times \mathbf{FM}_{|S_1|}(\mathbb{C}) \times \dots \times \mathbf{FM}_{|S_r|}(\mathbb{C}) \longrightarrow \mathbf{FM}_k(\mathbb{C})$$

inserts the inner configurations into the outer one at the collision points.

- (ii) **Open-to-open.** For $m \geq 1$ open inputs and one open output:

$$\mathbf{SC}^{\text{ch},\text{top}}(\text{op}^m; \text{op}) = \mathbf{E}_1(m) = \text{Conf}_m^<(\mathbb{R}). \quad (2.2)$$

Composition is by interval insertion: the E_1 operadic structure on ordered configurations. Since $\text{Conf}_m^<(\mathbb{R})$ is contractible, the open-to-open component is equivalent to the associative operad Ass up to homotopy.

- (iii) **Mixed (closed and open inputs, open output).** For $k \geq 0$ closed inputs and $m \geq 1$ open inputs with one open output:

$$\text{SC}^{\text{ch,top}}(\text{cl}^k, \text{op}^m; \text{op}) = \text{FM}_k(\mathbb{C}) \times E_1(m). \quad (2.3)$$

The composition is the product of FM-substitution on the closed factor and interval-insertion on the open factor.

- (iv) **Directionality (open-to-closed is empty).** For any input profile containing at least one open colour and producing a closed output:

$$\text{SC}^{\text{ch,top}}(\dots, \text{op}, \dots; \text{cl}) = \emptyset. \quad (2.4)$$

Bulk restricts to boundary; boundary does not act on bulk.

Remark 2.4 (Directionality and the Springer pattern). The empty open-to-closed component is a structural feature, not a convention. In the Beilinson–Drinfeld framework, the factorisation D -module on $\text{Ran}(X)$ (the closed sector) restricts to the boundary via $\iota^!$, but the reverse extension $\iota_!$ produces zero !-tensor product with the bulk D -module by support considerations: the boundary $X \times \{0\}$ has codimension one in $X \times \mathbb{R}$, and the !-extension carries no sections transverse to the boundary.

Remark 2.5 (Comparison with the classical SC). Voronov’s classical Swiss-cheese operad SC has E_2 closed colour (configurations of little disks in a disk) and E_1 open colour (configurations of little intervals in an interval). The operad $\text{SC}^{\text{ch,top}}$ replaces the E_2 closed colour by $\text{FM}_k(\mathbb{C})$ -spaces that carry holomorphic structure: the complex structure on $\mathbb{C}$ is visible in the boundary strata of $\text{FM}_k(\mathbb{C})$ and enters the bar differential through the OPE residues. On cohomology, the Kontsevich formality theorem provides a quasi-isomorphism from $\text{FM}_k(\mathbb{C})$ -chains to the E_2 -operad; composing with the projection from SC to the homology operad (Com, Ass) recovers the classical descent to PVA. At the chain level, the holomorphic structure is load-bearing: it determines the spectral parameters of the A_∞ operations.

Definition 2.6 ($\text{SC}^{\text{ch,top}}$ -algebra). An $\text{SC}^{\text{ch,top}}$ -algebra is a pair (B, A) where:

- (i) A is an E_1 -algebra (the open/boundary algebra);
- (ii) B is an algebra over the FM-operad $\{\text{FM}_k(\mathbb{C})\}_{k \geq 1}$ (the closed/bulk algebra);
- (iii) the mixed operations $\text{FM}_k(\mathbb{C}) \times E_1(m) \rightarrow \text{Hom}(B^{\otimes k} \otimes A^{\otimes m}, A)$ define a module structure of B over A , compatible with all composition maps;
- (iv) the open-to-closed maps are zero.

Example 2.7 (The chiral derived centre pair). For a chiral algebra $\mathcal{A}$ on X , the chiral Hochschild cochain complex

$$C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}) := \text{Hom}_{\overline{B}(\mathcal{A})}(\overline{B}(\mathcal{A}), \mathcal{A})$$

(the convolution Hom from the bar resolution into $\mathcal{A}$) carries the structure of an algebra over $\{\text{FM}_k(\mathbb{C})\}_{k \geq 1}$ via the chiral brace operations. The pair $(C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}), \mathcal{A})$ is an $\text{SC}^{\text{ch}, \text{top}}$ -algebra. The closed colour is the bulk; the open colour is the boundary. The bar complex $\overline{B}(\mathcal{A})$ is used *in the construction* of $C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$ but does not itself carry the $\text{SC}^{\text{ch}, \text{top}}$ structure.

3. HOMOTOPY-KOSZULITY

3.1. Koszulity of the classical Swiss-cheese operad. Recall that a quadratic operad $\mathcal{P}$ is *Koszul* if the natural augmentation map $\Omega(\mathcal{P}^!) \xrightarrow{\sim} \mathcal{P}$ is a quasi-isomorphism, where $\mathcal{P}^!$ is the Koszul dual cooperad and Ω denotes the cobar functor. For coloured operads, the Koszul property is defined colour by colour with compatibility conditions on the mixed sectors.

Theorem 3.1 (Livernet [13]). *The classical Swiss-cheese operad SC , with E_2 closed colour and E_1 open colour, is Koszul.*

The proof proceeds by establishing a distributive law between the Com and Ass components that governs the mixed sector, then verifying the rewriting criterion of Dotsenko–Khoroshkin [4] for the resulting coloured quadratic presentation.

3.2. Formality of the closed colour. The closed colour of $\text{SC}^{\text{ch}, \text{top}}$ is the FM-operad $\{\text{FM}_k(\mathbb{C})\}_{k \geq 1}$. Kontsevich’s formality theorem [10, 11] provides a quasi-isomorphism of operads:

$$\{H_{\bullet}(\text{FM}_k(\mathbb{C}))\}_{k \geq 1} \xleftarrow{\sim} \{\text{FM}_k(\mathbb{C})\}_{k \geq 1}, \quad (3.1)$$

where the left-hand side is the homology operad, which is the E_2 -operad $\{H_{\bullet}(\text{Conf}_k(\mathbb{C}))\}$ concentrated in degree zero. The formality is over $\mathbb{R}$; over $\mathbb{C}$ it holds unconditionally. The key input is the propagator on $\text{FM}_2(\mathbb{C})$ and the vanishing of all obstructions via the Arnold relation.

Remark 3.2 (Formality is for the closed colour, not the full operad). Kontsevich formality applies to the closed colour of $\text{SC}^{\text{ch}, \text{top}}$ (the FM-operad on $\mathbb{C}$). It does *not* assert formality of the full two-coloured operad $\text{SC}^{\text{ch}, \text{top}}$; the mixed sector carries non-trivial chain-level data that does not simplify under the formality quasi-isomorphism. The chain-level mixed data is precisely the spectral-parameter dependence of the A_{∞} operations.

3.3. Transfer to $\text{SC}^{\text{ch}, \text{top}}$.

Theorem 3.3 (Homotopy-Koszulity of $\text{SC}^{\text{ch}, \text{top}}$). *The operad $\text{SC}^{\text{ch}, \text{top}}$ is homotopy-Koszul: the canonical map*

$$\Omega \text{B}(\text{SC}^{\text{ch}, \text{top}}) \xrightarrow{\sim} \text{SC}^{\text{ch}, \text{top}} \quad (3.2)$$

is a quasi-isomorphism. The bar-cobar adjunction on SC^{ch,top}-algebras is a Quillen equivalence.

Proof. The argument has three steps.

Step 1. Livernet's Theorem 3.1 gives Koszulity of the classical SC: the bar-cobar resolution $\Omega(\mathrm{SC}^!) \xrightarrow{\sim} \mathrm{SC}$ is a quasi-isomorphism.

Step 2. Kontsevich formality (3.1) provides a chain of quasi-isomorphisms connecting the closed colour of SC^{ch,top} (the FM-operad) to the closed colour of SC (the E₂-operad, the homology of FM). The open colour is already E₁ in both operads, so no formality argument is needed there.

Step 3. The bar-cobar transfer principle (Loday–Vallette [14], Theorem 11.3.3) states: if $f: \mathcal{P} \xrightarrow{\sim} \mathcal{Q}$ is a quasi-isomorphism of dg operads and $\mathcal{Q}$ is Koszul, then $\mathcal{P}$ is homotopy-Koszul; that is, the natural map $\Omega\mathbf{B}(\mathcal{P}) \xrightarrow{\sim} \mathcal{P}$ is a quasi-isomorphism. Applying this with $\mathcal{P} = \mathrm{SC}^{\mathrm{ch,top}}$ and $\mathcal{Q} = \mathrm{SC}$ (connected by the formality quasi-isomorphism on the closed colour and the identity on the open colour) yields the result.

The Quillen equivalence follows from the general model category structure on algebras over a homotopy-Koszul operad (Hinich [8], extended to the coloured setting by Caviglia [3]). $\square$

Remark 3.4 (Why homotopy-Koszulity matters). Homotopy-Koszulity of SC^{ch,top} has three consequences:

- (a) The bar construction $\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}})$ is a well-defined cooperad with $\partial^2 = 0$; this is the prerequisite for the convolution algebra $\mathrm{Hom}_\Sigma(\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}}), \mathrm{End}_{(B,A)})$ to carry an L_∞ -structure and for the MC element α_T to exist.
- (b) The cobar resolution $\Omega\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}})$ provides a cofibrant replacement of SC^{ch,top}; algebras over this resolution are SC^{ch,top}-algebras up to homotopy.
- (c) The homotopy transfer theorem applies: given a deformation retract of the pair (B, A) onto cohomology $(H^\bullet(B), H^\bullet(A))$, the SC^{ch,top}-algebra structure transfers to a homotopy SC^{ch,top}-algebra structure on cohomology. This transfer is the mechanism producing the PVA structure.

4. THE KOSZUL DUAL: $(\mathrm{SC}^{\mathrm{ch,top}})^!$ IS NOT SELF-DUAL

4.1. The Koszul dual cooperad.

Theorem 4.1 (Koszul dual of SC^{ch,top}). *The Koszul dual cooperad of SC^{ch,top} is*

$$(\mathrm{SC}^{\mathrm{ch,top}})^! = (\mathrm{Lie}^c, \mathrm{Ass}^c, \text{shuffle-mixed}), \quad (4.1)$$

with the following structure:

- (i) **Closed colour.** $(\mathrm{SC}^{\mathrm{ch,top}})^!_{\mathrm{cl}}(n) = \mathrm{Lie}^c(n)$, the Lie cooperad. The dimension is $\dim(\mathrm{SC}^{\mathrm{ch,top}})^!_{\mathrm{cl}}(n) = (n-1)!$.

- (ii) **Open colour.** $(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})_{\mathrm{op}}^!(m) = \mathrm{Ass}^c(m) = \mathbb{k}$, the associative cooperad, one-dimensional in each degree.
- (iii) **Mixed sector.** $(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})_{\mathrm{cl}}^!(\mathrm{cl}^k, \mathrm{op}^m; \mathrm{op})$ carries the shuffle cooperad structure:

$$\dim(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})_{\mathrm{cl}}^!(\mathrm{cl}^k, \mathrm{op}^m; \mathrm{op}) = (k-1)! \cdot \binom{k+m}{m}. \quad (4.2)$$

Proof. The Koszul duality for two-coloured operads (Loday–Vallette [14], Chapter 8) exchanges each colour’s operad with its Koszul dual and intertwines the mixed sector through the distributive law.

For the closed colour: $\mathrm{Com}^! = \mathrm{Lie}$ (the classical Ginzburg–Kapranov duality [7]). The operation spaces exchange: $\dim \mathrm{Com}(n) = 1$ becomes $\dim \mathrm{Lie}(n) = (n-1)!$.

For the open colour: $\mathrm{Ass}^! = \mathrm{Ass}$ (associative is self-dual under Koszul duality). The operation spaces are one-dimensional in each degree, and this is preserved.

For the mixed sector: the distributive law of Com over Ass dualises to the shuffle structure. The dimension formula (4.2) follows from the shuffle product: $(k-1)!$ from the Lie component and $\binom{k+m}{m}$ from the shuffle of closed and open inputs. $\square$

Warning 4.2 ($\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ is not Koszul self-dual). The closed-colour dimensions differ:

n	1	2	3	4	5	6
$\dim \mathrm{SC}_{\mathrm{cl}}^{\mathrm{ch},\mathrm{top}}(n) = \dim \mathrm{Com}(n)$	1	1	1	1	1	1
$\dim(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})_{\mathrm{cl}}^!(n) = \dim \mathrm{Lie}(n)$	1	1	2	6	24	120

At $n \geq 3$ these are manifestly different. The operad $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ is therefore *not* Koszul self-dual.

What *is* true: the Koszul duality *functor* on $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$ -algebras is an involution. For any Koszul operad $\mathcal{P}$, the double dual satisfies $(\mathcal{P}^!)^! \cong \mathcal{P}$ (tautologically). This means the duality functor $\mathcal{A} \mapsto \mathcal{A}^!$ on algebras is an involution; it does *not* mean the operad is isomorphic to its own dual. The confusion between “the functor is an involution” and “the operad is self-dual” is a common source of error.

4.2. Algebras over $(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})^!$.

Proposition 4.3 (Structure of an $(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})^!$ -algebra). *An algebra over $(\mathrm{SC}^{\mathrm{ch},\mathrm{top}})^!$ is a pair (L, A) where:*

- (i) L is a Lie algebra (the closed colour, from the Lie component);
- (ii) A is an associative algebra (the open colour, from the Ass component);
- (iii) L acts on A by derivations, through the mixed sector;
- (iv) the open-to-closed direction is zero.

The chiral Koszul dual $\mathcal{A}_{\mathrm{line}}^!$ of a chiral algebra carries exactly this structure: a Lie bracket (the Sklyanin bracket from the closed colour) and an associative

product (the Yangian product from the open colour), with the Lie bracket acting as derivations.

Proof. This is a direct consequence of Theorem 4.1. The Lie cooperad dualises to the Lie operad on algebras; the Ass cooperad dualises to the Ass operad on algebras; the shuffle mixed sector dualises to the derivation condition. $\square$

Remark 4.4 (The asymmetry principle). The Koszul dual exchanges Com (one-dimensional, commutative) with Lie $((n-1)!$ -dimensional, non-commutative) on the closed colour, while preserving Ass on the open colour. This asymmetry is the operadic origin of the structural difference between the bulk and the line side: the bulk is governed by Com (factorisation on $\text{Ran}(X)$ is symmetric), while the line-side Koszul dual is governed by Lie (the Sklyanin bracket, the Yangian Lie structure). The open colour, governed by Ass on both sides, is the shared associative structure: the E_1 -algebra on the boundary equals the E_1 -algebra on the line side.

5. PVA DESCENT: A_∞ OPERATIONS FROM $\text{FM}_n(\mathbb{C})$

5.1. The A_∞ operations. The bar differential $D_{\mathcal{A}}$ on $\overline{B}(\mathcal{A}) = T^c(s^{-1}\overline{\mathcal{A}})$ is a coderivation of the cofree coalgebra. By the universal property of cofree coalgebras, a coderivation is determined by its composition with the cogenerator projection $\text{pr}: T^c(s^{-1}\overline{\mathcal{A}}) \rightarrow s^{-1}\overline{\mathcal{A}}$. The resulting sequence of multilinear maps

$$m_k: \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1)) \cdots ((\lambda_{k-1})), \quad k \geq 1, \quad |m_k| = 2 - k, \quad (5.1)$$

are the A_∞ operations. Here $\lambda_j = z_j - z_k$ are spectral parameters recording the holomorphic separations of the insertion points on X .

The identity $D_{\mathcal{A}}^2 = 0$ expands into the *Stasheff relations*: for each $n \geq 1$,

$$\sum_{\substack{i+j=n+1 \\ i,j \geq 1}} \sum_{s=0}^{n-j} (-1)^{\epsilon(s,j)} m_i(a_1, \dots, a_s, m_j(a_{s+1}, \dots, a_{s+j}), a_{s+j+1}, \dots, a_n) = 0, \quad (5.2)$$

where $\epsilon(s, j) = \sum_{t=1}^s (|a_t| - 1)$ is the Koszul sign from commuting m_j past the desuspended elements.

5.2. Stasheff relations as Stokes' theorem. Each Stasheff relation at degree n is Stokes' theorem on the Fulton–MacPherson space $\text{FM}_n(\mathbb{C})$. The boundary $\partial \text{FM}_n(\mathbb{C})$ is stratified by collision patterns: each codimension-one stratum D_S (indexed by a subset $S \subseteq \{1, \dots, n\}$ with $|S| \geq 2$) records the collision of the points indexed by S . The contribution of D_S to the Stasheff relation is $m_{n-|S|+1}$ composed with $m_{|S|}$, where $m_{|S|}$ acts on the colliding points and $m_{n-|S|+1}$ acts on the result together with the remaining points.

Proposition 5.1 (Stasheff = Stokes). *The n -th Stasheff relation (5.2) is the identity*

$$\int_{\text{FM}_n(\mathbb{C})} d\omega_n = \int_{\partial \text{FM}_n(\mathbb{C})} \omega_n = \sum_{|S| \geq 2} \int_{D_S} \omega_n,$$

where ω_n is the chain-level integrand of m_n contracted with the weight forms $\eta_{ij} = d \log(z_i - z_j)$ and OPE coefficients. The identity $\partial^2 = 0$ on $\text{FM}_n(\mathbb{C})$ (each codimension-two face is reached by exactly two codimension-one faces with opposite orientations) ensures $D_{\mathcal{A}}^2 = 0$.

Proof. The bar differential is defined as a sum of boundary integrals on $\text{FM}_n(\mathbb{C})$. The map m_n at degree n is the integral over the top-dimensional stratum of $\text{FM}_n(\mathbb{C})$; $m_i \circ m_j$ contributions arise from boundary strata. Stokes' theorem identifies the sum of boundary contributions with the differential of the bulk integral. The identity $d^2 = 0$ on differential forms, applied to the FM compactification (where boundary strata compose correctly), gives $D_{\mathcal{A}}^2 = 0$. $\square$

We spell out the first four cases.

5.2.1. $n = 1$: *the BRST differential.* $\text{FM}_1(\mathbb{C})$ is a single point with empty boundary. The Stasheff relation is $m_1 \circ m_1 = 0$: the map $m_1 = Q$ is a differential, $Q^2 = 0$. This is the BRST condition.

5.2.2. $n = 2$: *the Leibniz rule.* $\text{FM}_2(\mathbb{C})$, modulo translations, is a point; its boundary is the single collision stratum $z_1 = z_2$. The Stasheff relation is

$$m_1(m_2(a, b)) + m_2(m_1(a), b) + (-1)^{|a|} m_2(a, m_1(b)) = 0 :$$

the binary product m_2 is a chain map with respect to m_1 .

5.2.3. $n = 3$: *homotopy associativity.* The Stasheff associahedron K_3 is a closed interval $[0, 1]$, the quotient of $\text{FM}_3(\mathbb{C})$ by the affine group. Its two endpoints are the collision strata $D_{12} = \{z_1 \rightarrow z_2\}$ and $D_{23} = \{z_2 \rightarrow z_3\}$. The Stasheff relation is:

$$m_2(m_2(a, b), c) - m_2(a, m_2(b, c)) = m_1(m_3(a, b, c)) + m_3(m_1(a), b, c) + (-1)^{|a|} m_3(a, m_1(b), c) + (-1)^{|a|+|b|} m_3(a, b, m_1(c)) \quad (5.3)$$

The left-hand side is the associativity defect of m_2 . The right-hand side is the boundary of the homotopy m_3 : the binary product is associative up to the homotopy m_3 , which is the integral over the interval K_3 .

When $m_3 = 0$ (Heisenberg, free fermion), the relation reduces to strict associativity. When $m_3 \neq 0$ (affine Kac–Moody, Virasoro), the homotopy is the Jacobiator on K_3 .

5.2.4. $n = 4$: *the Stasheff pentagon*. The associahedron K_4 is a compact polygon with five boundary faces, corresponding to the collision strata D_{12} , D_{23} , D_{34} , D_{123} , D_{234} . The Stasheff relation involves m_2 , m_3 , and m_4 , with m_4 as the integral over the interior of K_4 . For $\mathcal{W}_3$, the operation m_4 is non-zero, with the coefficient $32/(22 + 5c)$ entering through the WWA channel (Section 7, third computation).

5.3. PVA descent on cohomology.

Definition 5.2 (Poisson vertex algebra). A *Poisson vertex algebra* (PVA) over $\mathbb{k}$ is a commutative differential algebra $(V, \cdot, \partial)$ equipped with a λ -bracket $\{a_\lambda b\} \in V[\lambda]$ satisfying the following five axioms:

- (PVA1) **Sesquilinearity.** $\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}$ and $\{a_\lambda \partial b\} = (\lambda + \partial)\{a_\lambda b\}$.
- (PVA2) **Skew-symmetry.** $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$.
- (PVA3) **Jacobi identity.** $\{a_\lambda\{b_\mu c\}\} - \{b_\mu\{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}$.
- (PVA4) **Left Leibniz rule.** $\{a_\lambda bc\} = \{a_\lambda b\}c + b\{a_\lambda c\}$.
- (PVA5) **Commutativity.** $a \cdot b = b \cdot a$.

Theorem 5.3 (PVA descent). *Let $\mathcal{A}$ be a chiral algebra on a smooth curve X , and let $H = H^\bullet(\mathcal{A}, m_1)$ be the cohomology with respect to the internal differential $m_1 = Q$. Suppose (H, p, ι, h) is a deformation retract of $(\mathcal{A}, m_1)$ onto H , with the side conditions $h^2 = h\iota = ph = 0$.*

Then:

- (a) *The homotopy transfer of the A_∞ structure $\{m_k\}$ to H produces transferred operations $\{m_k^H\}_{k \geq 1}$ satisfying the Stasheff relations.*
- (b) *The transferred binary operation m_2^H decomposes as*

$$m_2^H(a, b; \lambda) = a \cdot b + \{a_\lambda b\},$$

where $a \cdot b = m_2^H(a, b; 0)$ is a commutative associative product on H and $\{a_\lambda b\}$ is a λ -bracket.

- (c) *The pair $(H, \cdot, \{-\lambda-\})$ is a Poisson vertex algebra in the sense of Definition 5.2.*
- (d) *The higher transferred operations m_k^H for $k \geq 3$ encode the homotopy corrections to the PVA structure; they vanish if and only if $\mathcal{A}$ is formal (class G in the shadow depth classification).*

Proof outline. **Part (a)** follows from the homotopy transfer theorem for A_∞ algebras (Kadeishvili [9], Merkulov [15]), applied through the homotopy-Koszulity of SC^{ch,top} (Theorem 3.3).

Part (b). The binary operation m_2 has two sources: the constant term in λ (the normally-ordered product, which is commutative on cohomology) and the polynomial part in λ (the λ -bracket, which records the OPE singular part). On cohomology, the constant term survives as a commutative product because the Σ_n -equivariance of the chiral algebra (for E_∞ -chiral algebras, all standard vertex algebras) forces $m_2^H(a, b; 0) = m_2^H(b, a; 0)$.

Part (c). The five PVA axioms are the cohomological shadows of the Stasheff relations:

- Sesquilinearity (PVA1) is the $n = 2$ Stasheff relation applied to ∂a and a , using the translation covariance of the chiral algebra.
- Skew-symmetry (PVA2) follows from the Σ_2 -equivariance of $\text{FM}_2(\mathbb{C})$ (the antipodal map $z \mapsto -z$ on the boundary circle).
- The Jacobi identity (PVA3) is the $n = 3$ Stasheff relation (5.3) on cohomology, where m_3 disappears (it is exact on H) and the associativity defect becomes the Jacobiator.
- The Leibniz rule (PVA4) is the compatibility of m_2 with the product structure, inherited from the factorisation property of the D -module.
- Commutativity (PVA5) is the cohomological consequence of E_∞ -locality, as in part (b).

Part (d). If $\mathcal{A}$ is formal (all $m_k = 0$ for $k \geq 3$), then m_2 is strictly associative and the PVA structure on H is the complete algebraic datum. If $\mathcal{A}$ is non-formal (class L, C, or M), the higher operations m_k^H carry the residual A_∞ data that the PVA structure does not capture. $\square$

5.4. The five PVA axioms from geometry. We record the geometric origin of each PVA axiom in detail.

Proposition 5.4 (Geometric origin of PVA axioms). *The five PVA axioms correspond to geometric properties of the Fulton–MacPherson spaces as follows:*

<i>Axiom</i>	<i>FM property</i>	<i>Stasheff</i>
<i>Sesquilinearity</i>	<i>Translation covariance of $\text{FM}_2(\mathbb{C})$</i>	$n = 2$
<i>Skew-symmetry</i>	Σ_2 - <i>action on $\text{FM}_2(\mathbb{C})$</i>	$n = 2$
<i>Jacobi</i>	<i>Arnold relation on $\text{FM}_3(\mathbb{C})$</i>	$n = 3$
<i>Leibniz</i>	<i>Factorisation on $\text{FM}_2(\mathbb{C}) \times \text{FM}_1(\mathbb{C})$</i>	$n = 2$
<i>Commutativity</i>	E_∞ - <i>locality (Σ_n-equivariance)</i>	<i>all n</i>

Proof. Sesquilinearity. The translation $z \mapsto z + c$ acts on $\text{FM}_n(\mathbb{C})$ and preserves all boundary strata. On the bar complex, translation covariance is encoded by $D_{\mathcal{A}} \circ \partial = \partial \circ D_{\mathcal{A}}$, where ∂ is the translation operator on $\mathcal{A}$. Applied to a degree-two bar element $[s^{-1}\partial a \mid s^{-1}b]$, the Stasheff relation gives $m_2(\partial a, b; \lambda) = -\lambda m_2(a, b; \lambda)$ after extracting the λ -dependent part (the spectral parameter λ is conjugate to translation, so $\partial \leftrightarrow -\lambda$ under the Fourier-type correspondence between position and spectral variables). The right sesquilinearity follows by skew-symmetry.

Skew-symmetry. The involution $(z_1, z_2) \mapsto (z_2, z_1)$ on $\text{FM}_2(\mathbb{C})$ acts on the boundary circle $E_{12} \cong S^1$ as the antipodal map. On the bar complex, this exchanges $[s^{-1}a \mid s^{-1}b]$ with $(-1)^{(|a|-1)(|b|-1)}[s^{-1}b \mid s^{-1}a]$. On the λ -bracket, the antipodal map sends λ to $-\lambda - \partial$ (the shift by ∂ arises from the chain rule in the Fourier transform), yielding $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$.

Jacobi identity. The Arnold relation on $\mathrm{FM}_3(\mathbb{C})$,

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0, \quad (5.4)$$

where $\eta_{ij} = d \log(z_i - z_j)$, is the fundamental cohomological relation of the ordered configuration space. The $n = 3$ Stasheff relation on cohomology (where m_3 vanishes) reduces to

$$m_2(m_2(a, b), c) - m_2(a, m_2(b, c)) = 0 \quad (\text{modulo } m_3\text{-exact terms}).$$

Expanding m_2 into its product and λ -bracket components, the λ -bracket part of this relation is exactly the PVA Jacobi identity. The Arnold relation ensures the three cyclic terms cancel; this is the same mechanism that gives $D_{\mathcal{A}}^2 = 0$ at degree three.

Leibniz rule. The factorisation property of the chiral algebra gives a compatibility between the binary chiral product and the normally-ordered product. On $\mathrm{FM}_2(\mathbb{C}) \times \mathrm{FM}_1(\mathbb{C})$ (two colliding points and one spectator), the boundary stratum where the spectator approaches one of the colliding pair decomposes the interaction into two terms: $\{a_\lambda b\}c$ and $b\{a_\lambda c\}$. This is the Leibniz rule.

Commutativity. For $\mathbf{E}_\infty$ -chiral algebras (all standard vertex algebras), the chiral product is Σ_n -equivariant: the factorisation D -module on $\mathrm{Ran}(X)$ is defined on *unordered* configurations. On cohomology, this forces the normally-ordered product to be commutative: $a \cdot b = b \cdot a$. For genuinely $\mathbf{E}_1$ -chiral algebras (Yangians, Etingof–Kazhdan quantum vertex algebras), commutativity fails and the PVA axiom must be replaced by a noncommutative variant. $\square$

6. THE RESOLVENT PRINCIPLE

The A_∞ operations $\{m_k\}_{k \geq 2}$ are not independent. They are determined by a single datum: the binary product m_2 .

6.1. The tree-level bootstrap equation. Let (H, p, ι, h) be a deformation retract of $(\mathcal{A}, m_1)$ onto cohomology $H = H^\bullet(\mathcal{A}, m_1)$, with $h^2 = h\iota = ph = 0$.

Definition 6.1 (Dressed propagator). The *dressed propagator* $\Phi: T^+(H) \rightarrow \mathcal{A}$ is defined by the fixed-point equation

$$\Phi = \iota + h \circ m_2 \circ \Phi^{\otimes 2}. \quad (6.1)$$

Here $T^+(H) = \bigoplus_{k \geq 1} H^{\otimes k}$ is the reduced tensor module, $\iota: H \hookrightarrow \mathcal{A}$ is the inclusion, and $h: \mathcal{A} \rightarrow \mathcal{A}$ is the contracting homotopy.

Equation (6.1) is the tree-level bootstrap: the full propagator from boundary data H into the bulk algebra $\mathcal{A}$ is determined self-consistently from the free embedding ι and the interaction vertex m_2 . Iteration produces the tree formula.

6.2. The tree formula.

Theorem 6.2 (Resolvent principle). *The transferred A_∞ operations on H are given by*

$$m_k^H = \sum_{T \in \text{PRT}_k} p \circ \mathbf{m}(T) \circ \iota^{\otimes k}, \quad (6.2)$$

where:

- (i) PRT_k is the set of planar rooted binary trees with k leaves;
- (ii) $|\text{PRT}_k| = C_{k-1}$, the $(k-1)$ -st Catalan number ($C_0 = 1$, $C_1 = 1$, $C_2 = 2$, $C_3 = 5$, $\dots$);
- (iii) $\mathbf{m}(T)$ assigns the binary product m_2 to each internal vertex and the contracting homotopy h to each internal edge;
- (iv) $p: \mathcal{A} \rightarrow H$ is the projection and $\iota: H \rightarrow \mathcal{A}$ is the inclusion.

Proof. The iteration of (6.1) at order k collects all ways of building a k -to-1 map from $H^{\otimes k}$ to $\mathcal{A}$ using the vertex m_2 and the propagator h . Each such composition pattern is a planar rooted binary tree with k leaves (one leaf per input from H), internal vertices labelled by m_2 , and internal edges labelled by h . The projection p extracts the cohomology class.

The number of planar rooted binary trees with k leaves is the Catalan number C_{k-1} : this follows from the recursive decomposition of a tree into its left and right subtrees, which gives the recurrence $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$ with $C_0 = 1$.

The Stasheff relations for $\{m_k^H\}$ are verified by checking that the boundary contributions of each tree (where two adjacent vertices merge) produce the correct composition terms. This is the combinatorial content of the associahedron: the faces of K_n correspond to the degenerate trees. $\square$

One equation, one binary input, C_{k-1} trees.

Remark 6.3 (Physical interpretation). The tree formula is the perturbative expansion of the Lagrangian self-intersection $\mathcal{L} \times_{\mathcal{M}}^h \mathcal{L}$ in a shifted symplectic stack. Each tree is a Feynman diagram with vertices m_2 and propagators h . The sum over PRT_k is the tree-level scattering amplitude at multiplicity k . Loop corrections (genus $g \geq 1$) arise from the curvature $d_{\text{fb}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, which obstructs the tree-level bootstrap at higher genus.

7. THREE COMPUTATIONS

7.1. First computation: Heisenberg (formal, class G). Let $\mathcal{A} = \mathcal{H}_k$ be the Heisenberg algebra at level k , with generator J and OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}.$$

Proposition 7.1 (Heisenberg A_∞ structure). *The A_∞ structure of $\mathcal{H}_k$ is:*

- (i) $m_1 = 0$ (no internal differential).

- (ii) $m_2(J, J; \lambda) = k\lambda$ (the double-pole residue, absorbed by $d \log$: the OPE has a double pole, and the $d \log(z_1 - z_2)$ kernel reduces the pole order by one, producing a simple-pole collision residue k times the spectral parameter).
- (iii) $m_k = 0$ for all $k \geq 3$ (no composites, no higher poles).

The algebra is formal: the A_∞ structure is concentrated in m_2 .

Proof. The augmentation-ideal projected binary product $\bar{\mu}_2(J, J) = 0$: the OPE $J(z)J(w) \sim k/(z-w)^2$ has singular part a scalar (the identity), which projects to zero in $\bar{\mathcal{A}} = \ker(\varepsilon)$. Therefore the bar differential on $\bar{B}(\mathcal{H}_k)$ vanishes identically: all bar elements are $D_{\mathcal{A}}$ -closed.

The operation $m_2(J, J; \lambda) = k\lambda$ is obtained from the full OPE by the $d \log$ extraction: the coefficient of $1/(z_1 - z_2)^2$ in the OPE is k , and the spectral parameter $\lambda = z_1 - z_2$ after Fourier transform gives $k\lambda$.

Since $\bar{\mu}_2 = 0$ on $\bar{\mathcal{A}}$, the degree- n bar differential involves only the identity-valued OPE residue, which lives outside $\bar{\mathcal{A}}$. All m_k for $k \geq 3$ vanish. $\square$

Corollary 7.2 (Heisenberg PVA). *On $H^\bullet(\mathcal{H}_k) = \mathbb{k}[J]$, the PVA structure is:*

- (i) *Product: $J^n \cdot J^m = J^{n+m}$ (polynomial ring).*
- (ii) *λ -bracket: $\{J_\lambda J\} = k\lambda$.*
- (iii) *All higher operations $m_k^H = 0$ for $k \geq 3$.*

This is a PVA of rank one. The classical r -matrix is $r(z) = k/z$.

Verification: at $k = 0$, the r -matrix vanishes ($r(z) = 0$) and the algebra reduces to a commutative ring with trivial λ -bracket: the abelian limit.

The modular characteristic is $\kappa(\mathcal{H}_k) = k$: the same scalar governs the OPE, the λ -bracket, the r -matrix, and the genus-one curvature $d_{\text{fb}}^2 = k \cdot \omega_1$.

Remark 7.3 (Shadow depth zero). The Heisenberg algebra belongs to class G in the shadow depth classification: $d_{\text{alg}} = 0$. The shadow tower terminates at $\kappa = k$; the discriminant $\Delta = 8\kappa S_4 = 0$ (since $S_4 = 0$ for the Heisenberg) confirms the finite tower. The SC-formality condition is satisfied: $\mathcal{H}_k$ is SC-formal because $m_k = 0$ for $k \geq 3$.

7.2. Second computation: Kac–Moody at level k (Lie-transverse, class L). Let $\mathcal{A} = V_k(\mathfrak{g})$ be the vacuum module of the affine Kac–Moody algebra at level $k \in \mathbb{C} \setminus \{-h^\vee\}$, with generators $\{J^a\}$ ($a = 1, \dots, d = \dim \mathfrak{g}$) and OPE

$$J^a(z)J^b(w) \sim \frac{k\kappa^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}, \quad (7.1)$$

where κ^{ab} is the normalised Killing form and f^{ab}_c are the structure constants.

Proposition 7.4 (Kac–Moody A_∞ structure). *The A_∞ structure of $V_k(\mathfrak{g})$ is:*

- (i) $m_1 = 0$.

- (ii) $m_2(J^a, J^b; \lambda) = f^{ab}_c J^c + k \kappa^{ab} \lambda$. The simple-pole coefficient $f^{ab}_c J^c$ is the Lie bracket; the double-pole coefficient $k \kappa^{ab}$ contributes the λ -dependent term.
- (iii) $m_3 \neq 0$: the Jacobiator homotopy on $\text{FM}_3(\mathbb{C})$. The three codimension-one boundary strata of $\text{FM}_3(\mathbb{C})$ contribute $m_2(m_2(-, -), -)$ terms; their failure to cancel (the Jacobi identity defect from the simple-pole bracket) is corrected by m_3 .
- (iv) $m_k = 0$ for all $k \geq 4$, by dimension counting on $\text{FM}_k(\mathbb{C})$: the OPE has at most a double pole, so the chain-level integrands vanish for $k \geq 4$.

Proof. The binary operation extracts from the OPE (7.1): the residue at $z_1 = z_2$ with the $d \log(z_1 - z_2)$ kernel gives the simple-pole coefficient $f^{ab}_c J^c$ (the Lie bracket), and the double-pole coefficient $k \kappa^{ab}$ contributes the λ -linear term.

The existence of $m_3 \neq 0$ follows from the failure of strict associativity: $m_2(m_2(J^a, J^b), J^c) - m_2(J^a, m_2(J^b, J^c)) = f^{ab}_e f^{ec}_d J^d - f^{bc}_e f^{ae}_d J^d \neq 0$ (the Jacobi defect). The homotopy m_3 is the integral over $K_3 = [0, 1]$ that interpolates between the two boundary contributions.

The vanishing $m_k = 0$ for $k \geq 4$ follows from the pole structure: the OPE has at most double poles, and the weight forms η_{ij} absorb one pole order each. For $k \geq 4$ insertion points, the integrand on $\text{FM}_k(\mathbb{C})$ involves at most $(k - 1)$ weight forms but the pole excess is at most 2, which is consumed at $k = 3$. For $k \geq 4$, the integrand has no residual singularity and the FM integral vanishes. $\square$

Corollary 7.5 (Kac–Moody PVA). *On $H^\bullet(V_k(\mathfrak{g})) = \mathbb{k}[J^a : a = 1, \dots, d]$, the PVA structure is:*

- (i) *Product: polynomial ring in d generators.*
- (ii) *λ -bracket: $\{J^a_\lambda, J^b\} = f^{ab}_c J^c + k \kappa^{ab} \lambda$.*
- (iii) *Classical r -matrix: $r(z) = k \Omega / z$ in the trace-form convention, where $\Omega = \sum_a J^a \otimes J_a$ is the split Casimir.*

Verification: at $k = 0$, the r -matrix vanishes ($r(z) = 0$); the algebra reduces to the commutative ring with Lie bracket $\{J^a_\lambda, J^b\} = f^{ab}_c J^c$, which is still a well-defined PVA (the abelian limit of the level).

In the KZ convention: $r(z) = \Omega / ((k + h^\vee)z)$. Bridge: $k \Omega_{\text{tr}} = \Omega / (k + h^\vee)$ at generic k .

The modular characteristic is

$$\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}. \quad (7.2)$$

At $k = 0$: $\kappa = \dim(\mathfrak{g})/2$ (not zero: the pure vacuum anomaly). At $k = -h^\vee$: $\kappa = 0$ (critical level, the bar complex is flat).

Remark 7.6 (Shadow depth one, class L). The affine Kac–Moody algebra belongs to class L: $d_{\text{alg}} = 1$. The shadow tower extends to the cubic invariant $\mathfrak{C}(\hat{\mathfrak{g}}_k)$ (from the Lie structure constants) but terminates at shadow

depth $r_{\max} = 3$. The quartic shadow invariant $S_4 = 0$, so the discriminant $\Delta = 8\kappa S_4 = 0$: the shadow tower is finite (class L terminates at depth 3). The algebra is *not* SC-formal: $m_3 \neq 0$.

7.3. Third computation: $\mathcal{W}_3$ (infinite tower, class M). The $\mathcal{W}_3$ algebra has two generators: the stress tensor T of conformal weight 2 and a spin-3 current W of conformal weight 3.

Proposition 7.7 ($\mathcal{W}_3$ A_∞ structure). *The A_∞ structure of $\mathcal{W}_3$ satisfies:*

- (i) $m_1 = 0$.
- (ii) $m_2(T, T; \lambda) = (c/12)\lambda^3 + 2T\lambda + \partial T$ (the full TT OPE in divided-power parametrisation; the coefficient $c/12$ matches $T_{(3)}T = c/2$ via the divided power $\lambda^{(3)} = \lambda^3/3!$).
- (iii) $m_2(T, W; \lambda) = 3W\lambda + \partial W$.
- (iv) The WW operation involves the composite field $\Lambda = : (TT) : - \frac{3}{10}\partial^2 T$ at the λ -linear term:

$$m_2(W, W; \lambda) = \frac{c}{360}\lambda^5 + \frac{1}{3}T\lambda^3 + \frac{1}{2}(\partial T)\lambda^2 + \left(\frac{3}{10}\partial^2 T + \frac{32}{22+5c}\Lambda\right)\lambda + \dots$$

The coefficient $32/(22+5c)$ diverges at $c = -22/5$ (the Lee–Yang value).

- (v) $m_3 \neq 0$: the Jacobiator homotopy, involving Λ .
- (vi) $m_4 \neq 0$: the quartic operation from the $\text{FM}_4(\mathbb{C})$ boundary decomposition, with $32/(22+5c)$ entering through the $WW\Lambda$ channel.
- (vii) The A_∞ structure does not terminate: $m_k \neq 0$ for all $k \geq 3$.

Proof sketch. The binary operations are extracted from the Zamolodchikov $\mathcal{W}_3$ OPE [17] using the $d \log$ kernel and divided-power parametrisation. The non-vanishing of m_3 follows from the non-Lie structure of the $\mathcal{W}_3$ bracket (the bracket involves the composite field Λ , which is not a generator). The non-vanishing of m_4 follows from the Stasheff pentagon: the m_3 - m_3 composition on ∂K_4 does not cancel (the Λ channel provides a non-trivial residue on the interior of K_4). The infinite tower $m_k \neq 0$ for all k follows by induction: each m_k generates a non-vanishing correction at degree $k+1$ through the Stasheff cascade, driven by the non-primitive composite Λ . $\square$

Corollary 7.8 ($\mathcal{W}_3$ PVA). *On cohomology, the PVA structure is the Zamolodchikov PVA:*

$$\begin{aligned} \{T_\lambda T\} &= \frac{c}{12}\lambda^3 + 2T\lambda + \partial T, \\ \{T_\lambda W\} &= 3W\lambda + \partial W, \\ \{W_\lambda W\} &= \frac{c}{360}\lambda^5 + \frac{1}{3}T\lambda^3 + \frac{1}{2}(\partial T)\lambda^2 + \left(\frac{3}{10}\partial^2 T + \frac{32}{22+5c}\Lambda\right)\lambda + \dots \end{aligned}$$

The modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$: from the formula $\kappa(\mathcal{W}_N) = c(H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$, at $N = 3$ we have $H_3 = 1 + 1/2 + 1/3 = 11/6$, so $\kappa = c \cdot 5/6$.

Verification at $N = 2$: $H_2 = 3/2$, $H_2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2$, which matches the Virasoro formula $\kappa(\text{Vir}_c) = c/2$ (since $\mathcal{W}_2 = \text{Vir}$).

Remark 7.9 (Class M: infinite shadow depth). The $\mathcal{W}_3$ algebra belongs to class M: $d_{\text{alg}} = \infty$. The shadow tower $S_2, S_3, S_4, \dots$ is infinite, driven by the non-primitive composite Λ . The generating depth $d_{\text{gen}} = 3$ (the operation m_3 generates all higher m_k via the Stasheff cascade), but the algebraic depth is infinite (all $m_k \neq 0$). The discriminant $\Delta = 8\kappa S_4 \neq 0$ for $c \neq 0$ and $c \neq -22/5$, confirming the infinite tower.

This infinite depth is the algebraic shadow of the fact that three-dimensional gravity with $\mathcal{W}_3$ symmetry requires an infinite number of graviton vertices: the A_∞ structure is the perturbative expansion of the bulk theory.

Algebra	Class	d_{alg}	κ	$m_{k \geq 3}$	PVA type
$\mathcal{H}_k$	G	0	k	$= 0$	free rank-1
$V_k(\mathfrak{g})$	L	1	$\frac{d(k+h^\vee)}{2h^\vee}$	$m_3 \neq 0, m_{\geq 4} = 0$	affine, Lie-transverse
$\mathcal{W}_3$	M	∞	$5c/6$	all $\neq 0$	Zamolodchikov, infinite

8. WAVE-14 RECONCEPTION: HEPTAGON, TOPOLOGIZATION SCOPE, PVA DESCENT

This addendum reconceives the operad $\text{SC}^{\text{ch}, \text{top}}$ against three 2026-04-17 platonic-ideal anchors: the Vol II $\text{SC}^{\text{ch}, \text{top}}$ heptagon (seven edges routing bar, cobar, Drinfeld centre, derived centre, Swiss-cheese, PTVV, celestial), the four-class topologisation scope tree, and the Khan–Zeng PVA descent for freely-generated Poisson vertex algebras.

Theorem 8.1 (The $\text{SC}^{\text{ch}, \text{top}}$ heptagon). *The two-coloured operad $\text{SC}^{\text{ch}, \text{top}}$ sits at the hub of a heptagon of seven equivalent presentations, each edge a named theorem:*

- (1) Operadic hub: $\text{SC}^{\text{ch}, \text{top}}$ itself (this paper).
- (2) Koszul dual: $(\text{SC}^{\text{ch}, \text{top}})^! = (\text{Lie}^c, \text{Ass}^c, \text{shuffle-mixed})$ (this paper, Theorem 4.1; not Koszul self-dual).
- (3) Chiral factorization: the Beilinson–Drinfeld factorization algebra on $\text{Ran}(X)$ with open -decoration.
- (4) BV–BRST: the Costello–Gwilliam bulk–boundary BV complex.
- (5) Convolution: the L_∞ -algebra $\text{Hom}_\alpha(C, A)$ with open-closed decoration.
- (6) Drinfeld centre: $Z(\text{fact}(\mathcal{A})) \simeq_{\text{fact}} (Z_{\text{ch}}^{\text{der}}(\mathcal{A}))^{E_2}$ via categorified bar–cobar with half-braiding.
- (7) PTVV derived-AG: the 1-shifted symplectic mapping stack $\text{Map}(X \times_{\geq 0}, B \text{SC-Alg})$.

Each of the seven is a distinct theorem; the Vol II heptagon chapter (`sc_chtop_heptagon.tex`) proves the seven edges as named theorems.

Theorem 8.2 (Topologization scope tree, AP-TOPOLOGIZATION). *The passage $\text{SC}^{\text{ch,top}} \rightarrow E_3^{\text{top}}$ via Sugawara topologisation branches by shadow class:*

- *Class G (Heisenberg): E_3^{top} immediate; commutative input, Dunn on F^{H_k} .*
- *Class L ($V_k(\mathfrak{g})$, $k \neq -h^\vee$): E_3^{top} proved on the original complex, via Costello–Gwilliam factorization on $\times$ and explicit BRST primitives $\eta_1^{(i)}, \eta_1^{(ii)}$ giving $[Q, \tilde{G}_1] = T_{\text{Sug}}$ at strict chain level.*
- *Class C ($\beta\gamma$, bc): E_3^{top} via Friedan–Martinec–Shenker bosonisation reducing to class G.*
- *Class M ($\cdot$, $\mathcal{W}_N$): E_3^{top} weight-completed, via half-BRST plus MC5 pattern; original-complex chain-level is the sole remaining open direction.*
- *Critical level $k = -h^\vee$: E_2^{top} (a dimension drop, not a failure; Sugawara is undefined at critical level).*

Without a conformal vector (or at critical level), the theory is stuck at $\text{SC}^{\text{ch,top}}$; this is the generic situation (Heisenberg, lattice, E_1 -chiral Yangian input, free fields without stress tensor), for which $\text{SC}^{\text{ch,top}}$ is the final answer and E_3 the special case.

Theorem 8.3 (Khan–Zeng PVA descent on freely generated PVAs). *Let A be a freely-generated Poisson vertex algebra with conformal vector. Then the A_∞ operations $\{m_k\}$ extracted by cogenerator projection from the bar differential of the associated graded $\text{gr}_{L_i}(A)$ descend to the five PVA axioms (sesquilinearity, skew-symmetry, Jacobi, Leibniz, commutativity) on cohomology. The descent covers the 3d Poisson sigma-model class of Khan–Zeng, comprising all freely-generated PVAs with conformal vector; in particular, it recovers the Vol II orbifold route via $/n$ -invariants preserving E_n -structure.*

Remark 8.4 (AP-SC-BAR discipline reaffirmed). As stated in Warning 1.1, the bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an E_1 chiral coassociative coalgebra, not an $\text{SC}^{\text{ch,top}}$ -coalgebra. The $\text{SC}^{\text{ch,top}}$ structure lives on the derived centre pair $(C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \mathcal{A})$; the bar complex serves as a resolution used to compute $C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$, not as an object carrying $\text{SC}^{\text{ch,top}}$ -structure. The six-step chain of errors (FM25) is explicitly retired: (1) the bar differential is not a closed colour; (2) deconcatenation is not an open colour; (3) two structures on a single coalgebra do not produce two colours; (4) $\text{SC}^{\text{ch,top}}$ emerges in the output via $\text{Hom}(B(\mathcal{A}), \mathcal{A})$, not in the input; (5) E_1 -chiral $\neq E_1$ -topological (chiral lives on a curve, topological on $\text{Conf}_k(\cdot)$); (6) the Steinberg analogy is retired.

Remark 8.5 (AP-SC-NOT-SELFDUAL reaffirmed). $(\text{SC}^{\text{ch,top}})^\dagger \neq \text{SC}^{\text{ch,top}}$: closed-colour dimensions differ at $n \geq 3$ ($\dim \text{Com}(n) = 1$ vs $\dim \text{Lie}(n) = (n-1)!$). The duality functor is an involution; the operad is *not* self-dual. Livernet’s proof gives Koszulity, not self-duality.

Remark 8.6 (Epistemic tags). Theorem 8.1: the seven heptagon edges are *proved* in the Vol II heptagon chapter; this paper’s operadic hub (edges 1–3) is proved directly. Theorem 8.2: class G, L, C are unconditional on the original complex; class M is weight-completed (AP-TOPOLOGIZATION). Theorem 8.3: unconditional for freely-generated PVAs; the general class M case reduces to the weight-completed topologization.

REFERENCES

- [1] D. Ayala and J. Francis, *Factorization homology of topological manifolds*, J. Topol. **8** (2015), no. 4, 1045–1084.
- [2] A. Beilinson and V. Drinfeld, *Chiral Algebras*, Amer. Math. Soc. Colloq. Publ., vol. 51, AMS, 2004.
- [3] G. Caviglia, *The Quillen model structure for coloured operads*, arXiv:1401.6983, 2014.
- [4] V. Dotsenko and A. Khoroshkin, *Gröbner bases for operads*, Duke Math. J. **153** (2010), no. 2, 363–396.
- [5] A. De Sole and V. Kac, *Finite vs. affine W-algebras*, Jpn. J. Math. **1** (2006), no. 1, 137–261.
- [6] J. Francis and D. Gaitsgory, *Chiral Koszul duality*, Selecta Math. (N.S.) **18** (2012), no. 1, 27–87.
- [7] V. Ginzburg and M. Kapranov, *Koszul duality for operads*, Duke Math. J. **76** (1994), no. 1, 203–272.
- [8] V. Hinich, *Homological algebra of homotopy algebras*, Comm. Algebra **25** (1997), no. 10, 3291–3323.
- [9] T. V. Kadeishvili, *On the theory of homology of fiber spaces*, Uspekhi Mat. Nauk **35** (1980), no. 3, 183–188; Int. Math. Res. Not. (2003), no. 3, 141.
- [10] M. Kontsevich, *Operads and motives in deformation quantization*, Lett. Math. Phys. **48** (1999), no. 1, 35–72.
- [11] M. Kontsevich, *Deformation quantization of Poisson manifolds*, Lett. Math. Phys. **66** (2003), no. 3, 157–216.
- [12] H. Kajiura and J. Stasheff, *Open-closed homotopy algebra in mathematical physics*, J. Math. Phys. **47** (2006), 023506.
- [13] M. Livernet, *A rigidity theorem for pre-Lie algebras*, J. Pure Appl. Algebra **207** (2006), no. 1, 1–18.
- [14] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren Math. Wiss., vol. 346, Springer, 2012.
- [15] S. Merkulov, *Strong homotopy algebras of a Kähler manifold*, Internat. Math. Res. Notices (1999), no. 3, 153–164.
- [16] A. Voronov, *The Swiss-cheese operad*, Contemp. Math. **239** (1999), 365–373.
- [17] A. B. Zamolodchikov, *Infinite additional symmetries in two-dimensional conformal quantum field theory*, Theor. Math. Phys. **65** (1985), no. 3, 1205–1213.

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